

Group symbol: **3**

Team: **1**

Project title: **Resource Allocation App**

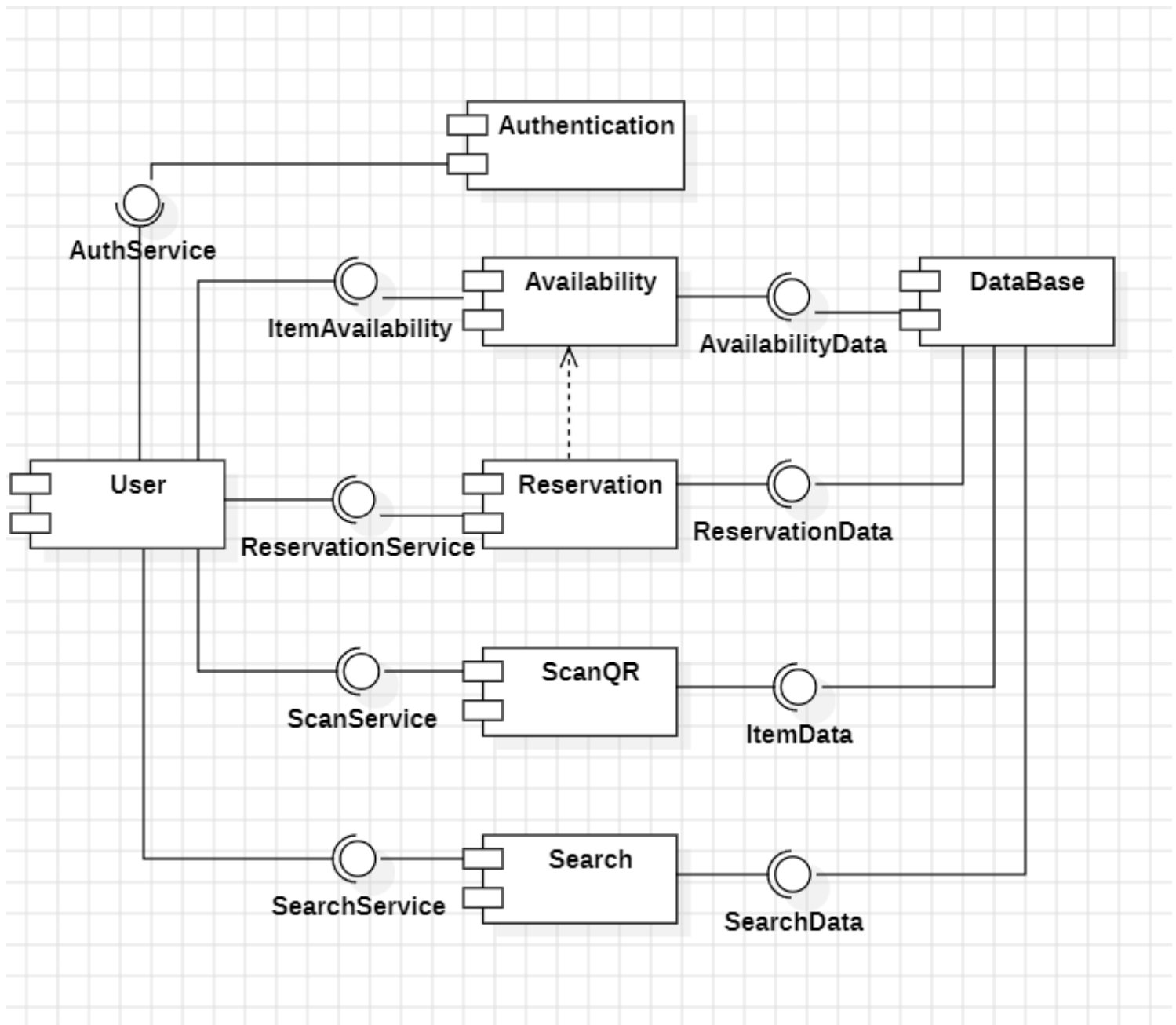
Team members (*filled by PM, Team Leader*):

No	Name	Surname	Student ID	Role
1	Carmelo Aidan	Romero	293936	<i>PM, Team Leader</i>
2	Álvaro	Puebla	293867	<i>Team member</i>
3	David	Sosa	293885	<i>Team member</i>
4	Diego	Ceballos	293943	<i>Team member</i>
5	Diego	García	293939	<i>Team member</i>
6	Óscar	López	293871	<i>Team member</i>

3. Design (S3)

3.1. Logical Software Architecture

The logical architecture aspect of the system should be present as a Component Diagram expressed in UML 2.5.

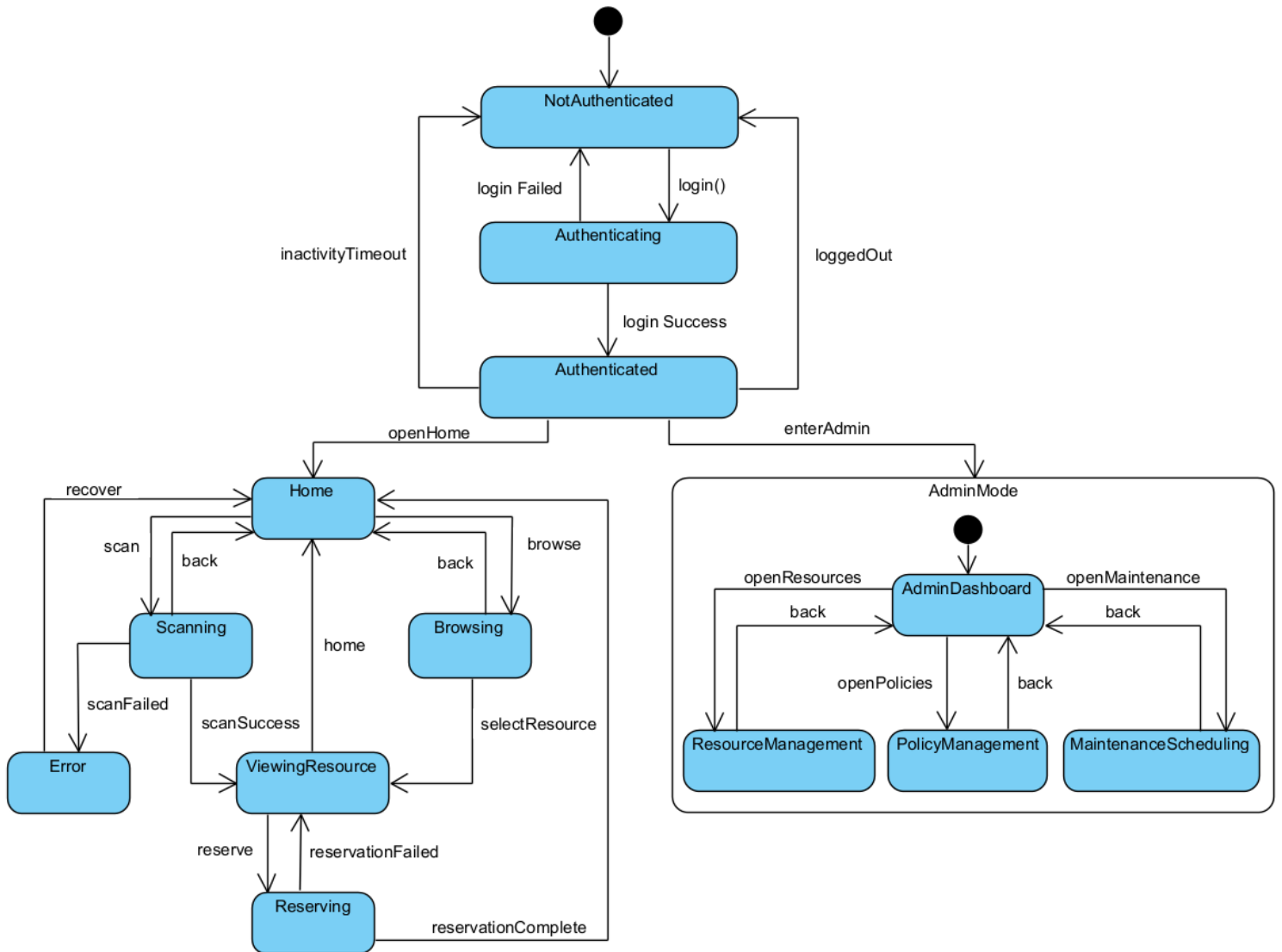


3.2. Business Logic Model

3.2.1. Behavioural Model

In this section present:

- state diagram for whole the system and



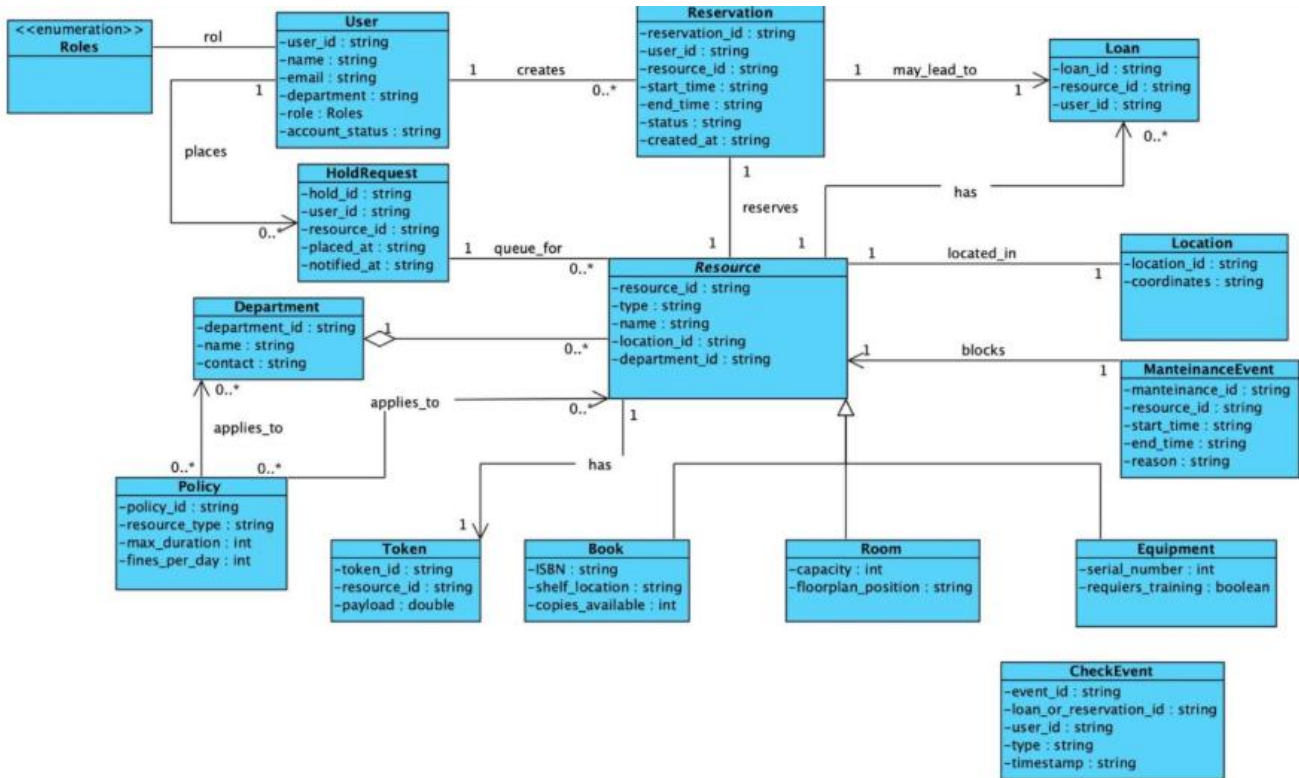
- *behaviour of 2 crucial use cases using the UML 2.5 diagrams, namely sequence diagram, and activity diagram.*

UC01	Making a reservation
Description	The client does a reservation for a book
Prerequisites	The client has logged in
Normal flow	
	1 The client provides the information needed to search the book.
	2 The system shows a list of books and their current availability.
Variant	
	1.2 The client scans the QR code of the book with the app.
	2.2 The system shows the book and its availability.
	3 The client selects a book to reserve
	4 The client makes the reservation.
	5 The system confirms the reservation

UC02	Cancelling a reservation
Description	The client cancels the reservation and returns the book.
Prerequisites	The client has logged in and has a reservation.
Normal flow	
	1 The client scans the QR code of the book with the app.
	2 The system shows the reservation.
	3 The client cancels the reservation and returns the book.
	4 The system confirms the cancellation and updates the book's availability.

3.2.2. Structural Model

Prepare the model as class diagram expressed in UML 2.5. Use natural language to describe the semantics of models' elements. Besides, use the object diagram to present an instance of the model at the time of system execution.



3.3. Database Model

3.3.1. Conceptual Model

The model contains the definition of entities and relationships among them in terms of the database (persistence). You should present it as a diagram and description of key elements in natural language.

User Attributes: • user_id (PK) • name • email • role (student / professor / admin)

Student Attributes: • student_id (PK, FK → User.user_id) • enrollment_number • degree_program

Professor Attributes: • professor_id (PK, FK → User.user_id) • department

Course Attributes: • course_id (PK) • name • professor_id (FK → Professor.professor_id)

Relationships: A professor teaches many courses; a course is taught by one professor.

Classroom Attributes: • classroom_id (PK) • name • capacity

ClassroomReservation Attributes: • reservation_id (PK) • classroom_id (FK → Classroom.classroom_id) • professor_id (FK → Professor.professor_id) • course_id (FK → Course.course_id) • date • start_time • end_time

Relationships: A classroom can have many reservations; a professor can make many reservations; each reservation belongs to one classroom, one professor, and one course.

Book Attributes: • book_id (PK) • title • author • year • available (boolean)

BookLoan Attributes: • loan_id (PK) • book_id (FK → Book.book_id) • student_id (FK → Student.student_id) • loan_date • return_date

Relationships: A student can borrow many books; a book can appear in many loans; each loan links one book and one student.

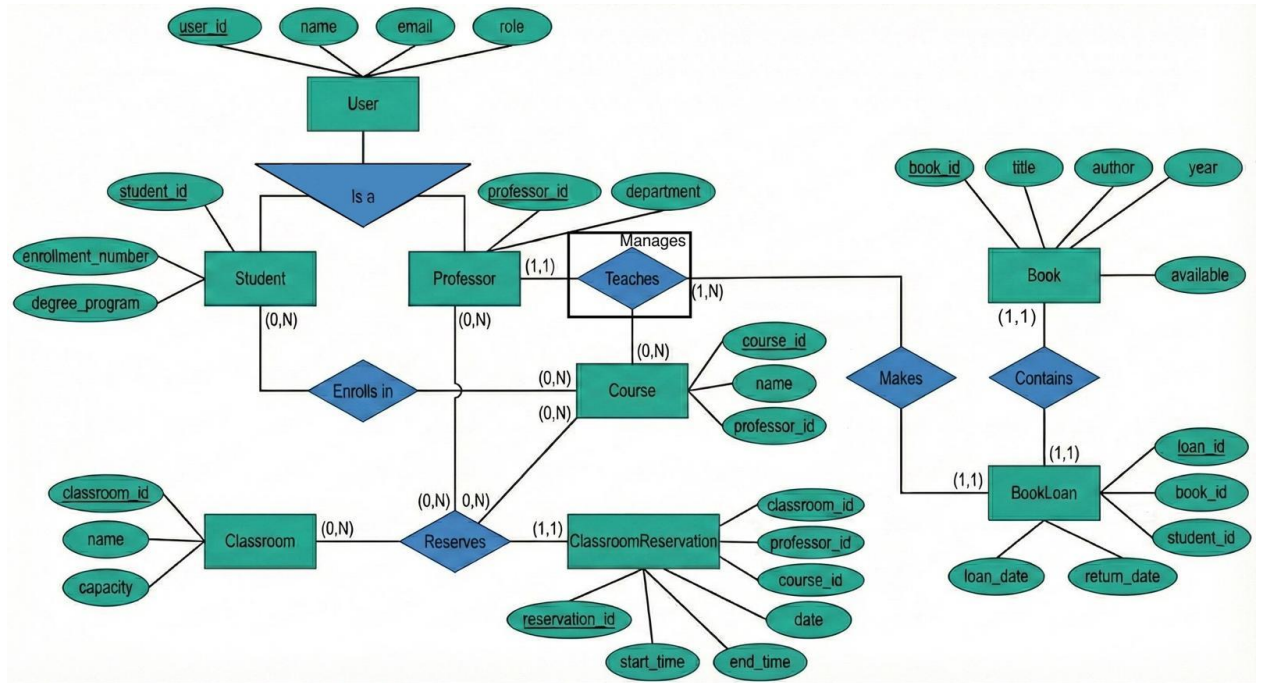
StudentCourse (optional, for enrollments) Attributes: • student_id (FK → Student.student_id) • course_id (FK → Course.course_id)

Relationship: Many-to-Many between students and courses.

3.3.2. Physical Model

The model presented as a diagram of the database that constitutes the implementation of the conceptual model in the selected database management system.

It is required to include an elaboration on the transformation of the conceptual model to the physical model. Description of transformation should include how elements like, e.g. inheritance and associations or other sophisticated modelling concepts have been handled.



3.4. User Interface Design

In this section, you should present the distribution of user interface components with events and their assignments to the behaviour specifications, which these elements initiate or take part.

These are the different pages we designed for this project, but we do not discard the fact that we may need to make new ones as the project advances and evolves.

